Practice Quiz 07C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

Find and repair a gap in a proof about relations.

1. A student claims that every symmetric, transitive relation R on a set A is reflexive, and argues:

   > Let $a \in A$. Choose $b \in A$ with aRb. Symmetry gives bRa. Transitivity then gives aRa.

   (a) Identify the unsupported step. Verify that $R = \{(1,1)\}$ on $A = \{1,2\}$ is a counterexample by checking symmetry, transitivity, and failure of reflexivity.

   Solution The unsupported step is choosing b with aRb: such a b may not exist. For example, take $R = \{(1,1)\}$ on $A = \{1,2\}$. It is symmetric because reversing its only pair changes nothing. It is transitive because the only possible two-step chain is $1R1$ and $1R1$, whose required conclusion $1R1$ holds. It is not reflexive on A, since $(2,2) \notin R$.

   (b) Prove that the claim becomes true if one also assumes $\forall a \in A\, \exists b \in A\, (aRb)$.

   Solution Suppose R is symmetric and transitive and satisfies the added hypothesis. Let $a \in A$ be arbitrary. The new hypothesis supplies $b \in A$ with aRb. By symmetry, bRa; applying transitivity to aRb and bRa gives aRa. Since this holds for every $a \in A$, R is reflexive on A. ∎

Solutions revised using GPT-6 Astra in the Codex harness.